THE MASTERMIND COACHING CLASSES

Revision Notes

Class 10 · Mathematics

Chapters 6 to 9 · NCERT Mathematics — Class 10 · CBSE / RBSE Syllabus 2026-27

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
6	Triangles	Similar figures, similarity criteria, basic proportionality theorem, areas of similar triangles, Pythagoras theorem
7	Coordinate Geometry	Distance formula, section formula and mid-point
8	Introduction to Trigonometry	Trigonometric ratios, values at standard angles, complementary angles, identities
9	Some Applications of Trigonometry	Heights and distances, angles of elevation and depression

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CHAPTER 06 · MATHS

Triangles

A photograph and its enlargement, a map and the land it shows, a pole and its shadow: each pair has the same shape but different sizes. Such figures are similar. This chapter studies similar triangles, the criteria that prove similarity, the powerful Basic Proportionality (Thales) theorem, the ratio of areas of similar triangles and the Pythagoras theorem with its converse.

Big Questions

- When are two figures, especially triangles, similar?
- What does a line parallel to one side of a triangle do to the other two sides?
- How are areas and sides of similar triangles related, and how does Pythagoras follow?

6.1 Similar Figures

DEFINITION

Two polygons with the same number of sides are **similar** if (i) their corresponding **angles are equal** and (ii) their corresponding **sides are in the same ratio (proportion)**. All circles of any radius are similar, as are all squares, but not all rectangles. Similar means the same shape; **congruent** figures are similar and also equal in size.

Two triangles ABC and DEF are similar (written triangle ABC ~ triangle DEF) if angle A = D, B = E, C = F and AB/DE = BC/EF = CA/FD. The order of vertices matters because it shows which parts correspond.

6.2 Basic Proportionality Theorem (Thales)

DEFINITION

BPT: a line drawn **parallel to one side** of a triangle, intersecting the other two sides at distinct points, divides those sides **in the same ratio**. If DE || BC in triangle ABC then **AD/DB = AE/EC**.

Idea of the proof. Join BE and CD. Triangles ADE and BDE have the same height from E, so their areas are in the ratio AD : DB; triangles ADE and CDE have the same height from D, so areas are in the ratio AE : EC. Triangles BDE and CDE have the same base DE and lie between the same parallels, so their areas are equal. Hence AD/DB = AE/EC.

REMEMBER

Converse of BPT: if a line divides two sides of a triangle in the same ratio, it is **parallel to the third side**. A **mid-point** line is parallel to the third side and half of it.

Example. DE || BC with AD = 3 cm, DB = 6 cm, AE = 4 cm: AE/EC = AD/DB = 1/2, so **EC = 8 cm**. **Example:** if AD/DB = 4/5 and AE/EC = 8/10 the line DE is parallel to BC, since both ratios equal 4/5.

6.3 Criteria for Similarity of Triangles

Criterion	Statement
AAA / AA	If two angles of one triangle equal two angles of another, the triangles are similar (third angles are then equal)
SSS	If the three sides of one triangle are proportional to the three sides of the other, they are similar
SAS	If one angle of a triangle equals one angle of another and the sides including these angles are proportional, the triangles are similar

Example. In triangle ABC with a point D on AB such that angle ACD = angle B, the triangles ACD and ABC share angle A and have angle ACD = angle B, so they are similar (AA), giving AC/AB = AD/AC, i.e. **AC^2 = AB x AD**.

Shadow problem. A pole 6 m high casts a shadow of 4 m while a tower casts 28 m at the same time. The sun's rays make equal angles, so the triangles are similar: 6/4 = h/28, **h = 42 m**.

6.4 Areas of Similar Triangles

FORMULA

If triangle ABC ~ triangle PQR, then **area(ABC)/area(PQR) = $(AB/PQ)^2$ = $(BC/QR)^2$ = $(CA/RP)^2$.** The ratio of areas equals the square of the ratio of corresponding sides, and also the squares of the ratios of corresponding altitudes or medians.

Example. Sides in ratio 3 : 5 give area ratio **9 : 25**. Two similar triangles of areas 81 cm^2 and 144 cm^2: if a side of the first is 6 cm, the corresponding side of the second is 6 x (12/9) = **8 cm**. If areas are in the ratio 4 : 9, medians are in the ratio 2 : 3.

6.5 Pythagoras Theorem

FORMULA

Pythagoras theorem: in a right triangle, the square of the hypotenuse equals the sum of the squares of the other two sides: **$AC^2 = AB^2 + BC^2$** (right angle at B). **Converse:** if in a triangle the square of one side equals the sum of the squares of the other two, the angle opposite the first side is a right angle.

Proof idea. The perpendicular from the right-angle vertex to the hypotenuse splits the triangle into two triangles similar to the whole; the proportional sides give AB^2 = AD x AC and BC^2 = CD x AC, which add to AC^2.

Sides	Check	Right triangle?
5, 12, 13	25 + 144 = 169	Yes
7, 24, 25	49 + 576 = 625	Yes
6, 8, 11	36 + 64 = 100 ≠ 121	No

Examples. A ladder 10 m long reaches a window 8 m above ground; its foot is √(100 − 64) = **6 m** from the wall. The diagonal of a rectangle 24 cm by 7 cm is √(576 + 49) = **25 cm**. A man goes 12 m west then 5 m north: he is **13 m** from the start. In an equilateral triangle of side a, altitude = (√3/2)a.

IMPORTANT

Match vertices carefully when writing similarity: write triangle ABC ~ triangle PQR only if A ↔ P, B ↔ Q, C ↔ R. The theorem on areas and the Pythagoras theorem are used as **results**; the proof of the theorem on areas of similar triangles is not asked.

Quick Revision: Likely Exam Questions

Likely question	Model answer
State BPT.	A line parallel to one side divides the other two sides in the same ratio.
If DE || BC, AD = 2, DB = 3, AC = 10, find AE.	AE/EC = 2/3, so AE = 4.
State the SAS similarity criterion.	One equal angle and proportional including sides.
Ratio of areas of similar triangles with sides 2 cm and 3 cm?	4 : 9.
Is a triangle with sides 9, 12, 15 right-angled?	Yes: 81 + 144 = 225.
Are all squares similar?	Yes.
Find the diagonal of a square of side 5 cm.	5 root 2 cm.
Which theorem proves a mid-point line is parallel to the third side?	Converse of BPT.

★ **KEY TAKEAWAYS**

- Similar figures: **equal angles and proportional sides**.
- **BPT:** DE || BC ⇒ AD/DB = AE/EC; its **converse** also holds.
- Similarity tests: **AA, SSS, SAS**.
- **Area ratio = (side ratio)^2**.
- **Pythagoras: $c^2 = a^2 + b^2$**; the converse identifies right triangles.

CHAPTER 07 · MATHS

Coordinate Geometry

A map reference, a seat number in a hall and a position on a screen are all described by coordinates. Coordinate geometry joins algebra with geometry: we place points on the plane and compute distances and positions with formulae. This chapter develops the distance formula and the section formula and uses them to identify shapes.

Big Questions

- How far apart are two points whose coordinates are known?
- How do we locate a point that divides a segment in a given ratio?
- How can coordinates prove a shape is a square, a rhombus or a parallelogram?

7.1 Cartesian Plane: Revision

The x-axis and y-axis meet at the **origin (0, 0)** and split the plane into four quadrants. A point is written (x, y); x is the **abscissa** and y the **ordinate**. A point on the x-axis is (x, 0); a point on the y-axis is (0, y).

Quadrant	Signs	Example
I	(+, +)	(3, 4)
II	(−, +)	(−3, 4)
III	(−, −)	(−3, −4)
IV	(+, −)	(3, −4)

7.2 Distance Formula

FORMULA

The distance between $P(x_1, y_1)$ and $Q(x_2, y_2)$ is $\mathbf{PQ = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}}$. The distance of P(x, y) from the origin is $\mathbf{\sqrt{(x^2 + y^2)}}$.

The formula comes from the Pythagoras theorem applied to the right triangle with PQ as hypotenuse and horizontal and vertical legs $|x_2 - x_1|$ and $|y_2 - y_1|$.

Examples. Distance between (2, 3) and (4, 1) is $\sqrt{(4 + 4)} = \mathbf{2\sqrt{2}}$. Distance from (−5, 7) to (0, 0) is $\sqrt{(25 + 49)} = \sqrt{74}$. **Point on the x-axis equidistant from (2, −5) and (−2, 9):** let it be (x, 0); $(x - 2)^2 + 25 = (x + 2)^2 + 81$ gives $-8x = 56$, so **x = −7**, point (−7, 0).

▸ Using the distance formula to identify figures

To show that the points form	Prove that
Collinear points	The sum of two distances equals the third (AB + BC = AC)
Isosceles triangle	Two sides equal
Equilateral triangle	All three sides equal

To show that the points form	Prove that
Right triangle	Sum of squares of two sides = square of the third
Parallelogram	Opposite sides equal (or diagonals bisect each other)
Rhombus	All four sides equal
Rectangle	Opposite sides equal and diagonals equal
Square	Four sides equal and diagonals equal

Example. A(1, 2), B(3, 4), C(5, 6): AB = BC = $2\sqrt{2}$ and AC = $4\sqrt{2}$ = AB + BC, so they are **collinear**. **Square:** (1, 7), (4, 2), (−1, −1), (−4, 4) have all sides $\sqrt{34}$ and both diagonals $\sqrt{68}$, so the figure is a **square**. **Centre of circle:** if (2, −3) and (−4, 5) are ends of a diameter, the centre is (−1, 1) and the radius is 5.

7.3 Section Formula

FORMULA

The point P dividing the segment joining A(x_1, y_1) and B(x_2, y_2) **internally in the ratio m : n** is **P = ((m x_2 + n x_1)/(m + n), (m y_2 + n y_1)/(m + n)). Mid-point (m = n):** ((x_1 + x_2)/2, (y_1 + y_2)/2).

Example. Point dividing (−3, 10) and (6, −8) in the ratio 2 : 1 is ((12 − 3)/3, (−16 + 10)/3) = **(3, −2)**. **Points of trisection** of (4, −1) and (−2, −3): ratios 1 : 2 and 2 : 1 give (2, −5/3) and (0, −7/3).

Find the ratio in which a line divides a segment

Take the ratio as k : 1 (or m : n), apply the section formula to the coordinate that is known (e.g. y = 0 for the x-axis) and solve for k.

Example

The x-axis divides (2, −3) and (5, 6): y = (6k − 3)/(k + 1) = 0 gives k = 1/2, so the ratio is **1 : 2** and the point is **(3, 0)**.

Finding the fourth vertex

In a parallelogram the diagonals bisect each other: equate the mid-points of the two diagonals.

Example (parallelogram). Vertices A(1, 2), B(4, 3), C(6, 6) and D(x, y): mid-point of AC = (3.5, 4) equals mid-point of BD = ((4 + x)/2, (3 + y)/2), so **D = (3, 5)**.

IMPORTANT

In the section formula the ratio **m : n** places the weight **crosswise**: m is multiplied with the coordinates of B and n with those of A. A negative k from the 'ratio of division' means the point divides **externally**, which is not required here.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Distance between (3, 4) and (−3, −4)?	10.
Mid-point of (4, 6) and (−2, 2)?	(1, 4).
Find y so that the distance between (2, y) and (10, −3) is 10.	64 + $(y + 3)^2$ = 100, y = 3 or −9.

Likely question	Model answer
Coordinates of the point on the y-axis at distance 5 from the origin?	(0, 5) or (0, −5).
Point dividing (1, 3) and (4, 6) in the ratio 1 : 2?	(2, 4).
Do (−1, −1), (2, 3), (8, 11) lie on a line?	Check 5 + 10 = 15 (distances): yes, collinear.
What is the distance formula based on?	The Pythagoras theorem.
Condition for a parallelogram using diagonals?	Diagonals bisect each other (same mid-point).

★ KEY TAKEAWAYS

- **$d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$.**
- **Section formula** for ratio m : n; **mid-point** is the average of coordinates.
- Identify **collinear points and shapes** by comparing distances.
- To find an unknown ratio, take **k : 1** and use the given coordinate.

CHAPTER 08 · MATHS

Introduction to Trigonometry

Builders, navigators and surveyors cannot measure the height of a tower or the width of a river directly; they use the relation between angles and sides of a right triangle. Trigonometry gives this relation through ratios of the sides. This chapter defines the six ratios, finds their values at 0, 30, 45, 60 and 90 degrees, links ratios of complementary angles and proves three basic identities.

Big Questions

- What are the six trigonometric ratios of an acute angle?
- What are the values at the standard angles and for complementary angles?
- How do the identities help us simplify and prove expressions?

8.1 Trigonometric Ratios

In a right triangle ABC with the right angle at B and an acute angle A, the side opposite A is the **perpendicular**, the side next to A is the **base** (adjacent), and the side opposite the right angle is the **hypotenuse**.

FORMULA

sin A = perpendicular / hypotenuse; cos A = base / hypotenuse; tan A = perpendicular / base = sin A / cos A. Reciprocals: **cosec A = 1/sin A; sec A = 1/cos A; cot A = 1/tan A = cos A / sin A.**

- The ratios depend only on the **angle**, not on the size of the triangle, because similar triangles have proportional sides.
- sin A and cos A are always **at most 1** (hypotenuse is the longest side); sec A and cosec A are **at least 1**; tan A can be any value.

Example. If tan A = 3/4: perpendicular 3, base 4, hypotenuse 5, so sin A = 3/5, cos A = 4/5. **If 15 cot A = 8:** cot A = 8/15 (base 8, perpendicular 15), hypotenuse 17: sin A = 15/17 and sec A = 17/8. **If cos A = 7/25**, then perpendicular = 24 and tan A = 24/7.

8.2 Ratios of Standard Angles

Angle	0°	30°	45°	60°	90°
sin	0	1/2	1/√2	√3/2	1
cos	1	√3/2	1/√2	1/2	0
tan	0	1/√3	1	√3	not defined
cosec	not defined	2	√2	2/√3	1
sec	1	2/√3	√2	2	not defined
cot	not defined	√3	1	1/√3	0

REMEMBER

For sin: write 0, 1, 2, 3, 4, divide by 4 and take the square root: 0, 1/2, 1/√2, √3/2, 1. Cos values are the same list in reverse order. As A increases from 0° to 90°, sin A rises from 0 to 1 and cos A falls from 1 to 0.

Examples. $\sin^2 30° + \cos^2 60° = 1/4 + 1/4 =$ **1/2**. $2\tan^2 45° + \cos^2 30° - \sin^2 60° = 2 + 3/4 - 3/4 =$ **2**. If $\sin(A - B) = 1/2$ and $\cos(A + B) = 1/2$ with $A > B$, then $A - B = 30°$ and $A + B = 60°$, so **$A = 45°, B = 15°$**.

8.3 Ratios of Complementary Angles

FORMULA

$\sin(90° - A) = \cos A$; $\cos(90° - A) = \sin A$; $\tan(90° - A) = \cot A$; $\cot(90° - A) = \tan A$; $\sec(90° - A) = \operatorname{cosec} A$; $\operatorname{cosec}(90° - A) = \sec A$.

Examples. $\sin 63°/\cos 27° = 1$. $\tan 26° \times \tan 64° = \tan 26° \times \cot 26° =$ **1**. $\cos 48° - \sin 42° = 0$. If $\sin 3A = \cos(A - 26°)$, then $3A + A - 26° = 90°$, so **$A = 29°$**.

8.4 Trigonometric Identities

FORMULA

$\sin^2 A + \cos^2 A = 1$; $1 + \tan^2 A = \sec^2 A$; $1 + \cot^2 A = \operatorname{cosec}^2 A$. Each is true for every angle for which the ratios are defined. They follow from Pythagoras in a right triangle (divide by hypotenuse^2, base^2 or perpendicular^2).

Useful forms	Expression
$\sin^2 A$	$1 - \cos^2 A = (1 - \cos A)(1 + \cos A)$
$\sec^2 A - \tan^2 A$	$1 = (\sec A - \tan A)(\sec A + \tan A)$
$\operatorname{cosec}^2 A - \cot^2 A$	1

Prove: $(1 - \cos\theta)(1 + \cos\theta)(1 + \tan^2\theta) = \tan^2\theta$

LHS $= \sin^2\theta \times \sec^2\theta = \sin^2\theta/\cos^2\theta = \tan^2\theta =$ RHS.

Prove: $\sec A\,(1 - \sin A)(\sec A + \tan A) = 1$

$(1 - \sin A)(1 + \sin A)/\cos^2 A = \cos^2 A/\cos^2 A = 1$.

Strategy

Start from the more complicated side, convert everything into sin and cos, use identities, and simplify to the other side.

Example. If $\sin\theta + \sin^2\theta = 1$ then $\cos^2\theta = \sin\theta$, so **$\cos^2\theta + \cos^4\theta = \sin\theta + \sin^2\theta = 1$**.

IMPORTANT

sin A is a single quantity; **sin A is not sin x A**. And sin(A + B) is not sin A + sin B. Do not mix degrees and radians; Class 10 uses degrees.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Define tan A in a right triangle.	Perpendicular divided by base (opposite/adjacent).
Value of sin 30° + cos 60°?	1.
If sin A = 3/5, find sec A.	5/4.

Likely question	Model answer
Evaluate $\sec^2 10° - \cot^2 80°$.	1.
What is tan 90°?	Not defined.
Simplify $(1 + \tan^2 A)(1 - \sin A)(1 + \sin A)$.	1.
Express sin A in terms of cos A.	$\sqrt{(1 - \cos^2 A)}$.
Value of cos 45° x sin 45°?	1/2.

★ KEY TAKEAWAYS

- **sin = P/H, cos = B/H, tan = P/B**; cosec, sec, cot are reciprocals.
- Learn the **table at 0, 30, 45, 60, 90 degrees**.
- **Complementary angles:** sin(90° − A) = cos A, etc.
- Three identities: **$\sin^2 + \cos^2 = 1$; $1 + \tan^2 = \sec^2$; $1 + \cot^2 = \text{cosec}^2$**.

CHAPTER 09 · MATHS

Some Applications of Trigonometry

How tall is the school flagpole? How far is a ship from a lighthouse? Instead of climbing or sailing, we measure an angle and one length and let trigonometry do the rest. This chapter applies the ratios of the previous chapter to problems on heights and distances using angles of elevation and depression.

Big Questions

- What are the line of sight, angle of elevation and angle of depression?
- How do we convert a word problem into a right triangle?
- How do we handle problems with two right triangles?

9.1 Key Terms

Term	Meaning
Line of sight	The line from the observer's eye to the object being viewed
Angle of elevation	Angle between the horizontal and the line of sight when the object is **above** the horizontal level (the observer looks up)
Angle of depression	Angle between the horizontal and the line of sight when the object is **below** the horizontal level (the observer looks down)

IMPORTANT

The angle of depression from a point A to an object B equals the angle of elevation from B to A, because they are **alternate angles** between parallel horizontals. Heights are measured **vertically** and distances **horizontally**, and the observer's height may have to be added.

9.2 Strategy for Problems

Step 1 — Draw a rough figure
Mark the observer, the object, the horizontal, the angles and given lengths.

Step 2 — Find the right triangle(s)
Name the sides as perpendicular (height), base (horizontal distance) and hypotenuse (line of sight, rope, ladder).

Step 3 — Choose the ratio
Use tan when the height and base are involved; sin when the hypotenuse and height are involved; cos for hypotenuse and base.

Step 4 — Solve and state the answer
Use root 2 = 1.41 and root 3 = 1.73 when asked, and give units.

Given	Required	Use
Height and base	Angle	tan
Angle and base	Height	height = base x tan

Given	Required	Use
Angle and hypotenuse	Height	height = hypotenuse x sin
Angle and height	Base	base = height / tan

FORMULA

Standard values: tan 30° = 1/√3, tan 45° = 1, tan 60° = √3; sin 30° = 1/2, sin 45° = 1/√2, sin 60° = √3/2.

9.3 Problems with One Right Triangle

- **Pole:** from a point 20 m from the foot of a tower, the angle of elevation of the top is 60°; height = 20 tan 60° = 20√3 = **34.6 m**.
- **Kite:** a kite string 100 m long is taut and makes 60° with the ground; height = 100 sin 60° = 50√3 = **86.6 m**.
- **Ladder:** a ladder 10 m long reaches a wall making 60° with the ground; it reaches 10 sin 60° = 5√3 = 8.66 m and its foot is 10 cos 60° = **5 m** from the wall.
- **Depression:** from the top of a 75 m lighthouse the angle of depression of a ship is 30°; distance = 75/tan 30° = 75√3 = **129.9 m**.
- **Shadow:** a tower 30 m high casts a shadow 30 m long; the sun's altitude is 45° since tan = 30/30 = 1. If the shadow is 10√3 m, tan θ = 30/(10√3) = √3 and θ = 60°.

9.4 Problems with Two Right Triangles

When the observer moves, or two objects are involved, two right triangles share a side (usually the vertical height). Express the common height or base in both triangles and equate.

Example 1. The angles of elevation of the top of a tower from two points on a line through its foot, at distances 40 m apart, are 30° and 60° (both on the same side). Let the height be h. Then the horizontal distances are h/tan 30° = h√3 and h/tan 60° = h/√3, and their difference is 40: h(√3 − 1/√3) = 40, so h x 2/√3 = 40 and **h = 20√3 = 34.6 m**.

Example 2 (building and tower). From the top of a building 30 m high the angle of elevation of the top of a tower is 30° and the angle of depression of its foot is 60°. Distance between them = 30/tan 60° = 10√3 m; the tower rises 10√3 x tan 30° = 10 m above the building, so tower height is **40 m**.

Example 3 (moving closer). A man walks 50 m towards a pole and the angle of elevation changes from 30° to 45°. With height h: h/tan 30° − h/tan 45° = 50, h(√3 − 1) = 50, so **h = 50/(0.73) = 68.5 m** (using √3 = 1.73).

Example 4 (bridge). From a helicopter hovering 200 m above a river, the angles of depression of the two ends of a straight bridge in line below it are 30° and 45°. The ends are 200 m and 200√3 m from the point below, so the bridge is 200(√3 − 1) = **146 m** long.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Define angle of elevation.	The angle between the horizontal and the line of sight to an object above the horizontal.
A pole 6 m high casts a 2√3 m shadow. Find the sun's elevation.	tan θ = 6/(2√3) = √3, so 60°.

Likely question	Model answer
At what angle does a ladder of 10 m reach 5 m height?	sin θ = 1/2, so 30°.
Angle of elevation of the sun when shadow equals the height?	45°.
Which ratio links height and base?	tan.
From a 50 m high tower the angle of depression of a car is 45°; how far is it?	50 m from the foot.
Why are depression and elevation equal?	Alternate angles between parallel horizontals.
What value of root 3 is used in calculation?	1.73 unless stated otherwise.

★ KEY TAKEAWAYS

- **Elevation** is measured upward and **depression** downward from the horizontal.
- Draw the figure, find the **right triangle**, choose **tan, sin or cos**.
- Two-triangle problems: express the **common height or base** twice and equate.
- Use **standard values** (√2 = 1.41, √3 = 1.73) and state units.